

Modern Beamer Presentations with the METROPOLIS package

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1 Introduction

Beamer is an awesome way to make presentations with LaTeX, but its theme selection is surprisingly sparse. The stock themes share an aesthetic that can be a little cluttered, while the few distinctive custom themes available are often specialized for a particular corporate or institutional brand.

The goal of **METROPOLIS** is to provide a simple, modern Beamer theme suitable for anyone to use. It tries to minimize noise and maximize space for content; the only visual flourish it offers is an (optional) progress bar added to each slide or to the section slides.

By default, **METROPOLIS** uses **Fira Sans**, a gorgeous typeface commissioned by Mozilla and designed by **Carrois**. For best results, you will need the Fira typeface installed and use $\text{X}_{\text{Y}}\text{TeX}$ to typeset your slides. However, **METROPOLIS** can also be used with other typefaces and $\text{L}^{\text{A}}\text{TeX}$ build systems.

METROPOLIS's codebase is maintained on **GitHub**. If you have issues, find mistakes in the manual or want to help make the theme even better, please get in touch there. The [full list of contributors](#) already contains over a dozen names!

2 Getting Started

2.1 Installing from CTAN

For most users, we recommend installing **METROPOLIS** from [CTAN](#). If you keep your T_EX distribution up-to-date, chances are good that **METROPOLIS** is already installed. If it is not, you need to update your packages. If your distribution is T_EX Live (or MacT_EX on OS X), the following command updates all packages.

```
tlmgr update --all
```

If this results in an error, you may need to run it with administrative privileges:

```
sudo tlmgr update --all
```

MacT_EX on OS X also provides a graphical interface for **tlmgr** called T_EX Live Utility.

For any other distribution please refer to its documentation on how to update your packages.

To get the most out of the theme you should also install the **Fira** fonts. However, this is not mandatory; **METROPOLIS** also works with the standard fonts.

2.2 Installing from GitHub

If you want to use the cutting-edge development version of **METROPOLIS**, you can install it manually. Like any T_EX package, this involves four easy steps:

Download the source with a `git clone` of the [METROPOLIS repository](#) or as a [zip archive](#) of the latest development version.

Compile the style files by running `make sty` inside the downloaded directory. (Or run T_EX directly on `source/metropolistheme.ins`.)

Move the resulting *.sty files to the folder containing your presentation. To use **METROPOLIS** with many presentations, run `make install` or move the *.sty files to a folder in your T_EX path instead.

Use the theme for your presentation by declaring `\usetheme{metropolis}` in the preamble of your Beamer document.

METROPOLIS uses the Make build system to offer the following installation options for advanced users:

`make sty` builds the theme style files.

`make doc` builds this documentation manual.

`make demo` builds a demo presentation to test the features of **METROPOLIS**.

`make all` builds the theme and manual.

`make clean` removes the files generated by `make all`.

`make install` installs the theme into your local texmf folder.

`make uninstall` removes the theme from your local texmf folder.

2.3 A Minimal Example

The following code shows a minimal example of a Beamer presentation using **METROPOLIS**.

```
\documentclass{beamer}
\usetheme{metropolis}      % Use metropolis theme
\title{A minimal example}
\date{\today}
\author{Matthias Vogelgesang}
\institute{Centre for Modern Beamer Themes}
\begin{document}
  \maketitle
  \section{First Section}
  \begin{frame}{First Frame}
    Hello, world!
  \end{frame}
\end{document}
```

2.4 Dependencies

METROPOLIS depends on the `beamer` class and the following standard packages:

- `tikz`
- `etoolbox`
- `ifxetex`
- `pgfopts`
- `calc`
- `ifluatex`

For best results, we recommend installing the fonts **Fira Sans** and **Fira Mono** and compiling with **METROPOLIS** using Xe_{La}TeX or Lua_{La}TeX. These are optional dependencies; **METROPOLIS** is compatible with (e.g.) pdf_{La}TeX and will fall back to standard fonts if **Fira Sans** or **Fira Mono** is not installed.

The packaged name of **Fira Sans** is **Fira Sans OT** in some Linux distributions; this case is automatically handled by **METROPOLIS**.

2.5 Pandoc

To use this theme with **Pandoc**-based presentations, you can run the following command

```
$ pandoc -t beamer --latex-engine=xelatex -V theme:
  metropolis -o output.pdf input.md
```

3 Customization

3.1 Package options

The theme provides a number of options, which can be set using a key=value interface. The primary way to set options is to provide a comma-separated list of option-value pairs when loading **METROPOLIS** in the preamble:

```
\usetheme[option1=value1, option2=value2, ...]{metropolis}
```

Options can be changed at any time — even mid-presentation! — with the `\metroset` macro.

```
\metroset{option1=newvalue1, option2=newvalue2, ...}
```

The list of options is structured as shown in the following example.

option key *list of possible values* default

A short description of the option.

3.1.1 Main theme

titleformat *regular, smallcaps, allsmallcaps, allcaps* regular

Changes the format of titles, subtitles, section titles, frame titles, and the text on “standout” frames. The available options produce Regular, SMALLCAPS, ALLSMALLCAPS, or ALLCAPS titles. Please refer to Section 6.1 for known issues with these options.

titleformat plain *regular, smallcaps, allsmallcaps, allcaps* regular

Changes the format of “standout” frames (see **titleformat**, above).

3.1.2 Inner theme

sectionpage *none, simple, progressbar* progressbar

Adds a slide at the start of each section (**simple**) with an optional thin progress bar below the section title (**progressbar**). The **none** option disables the section page.

subsectionpage *none, simple, progressbar* none

Optionally adds a slide at the start of each subsection. If enabled with the **simple** or **progressbar** options, the style of the **section page** will be updated to match the style of the **subsection page**. Note that section slides and subsection slides can appear consecutively if both are enabled; you may want to use this option together with **sectionpage=none** depending on the section structure of your presentation.

3.1.3 Outer theme

numbering	<i>none, counter, fraction</i>	counter
	Controls whether the frame number at the bottom right of each slide is omitted (none), shown (counter) or displayed as a fraction of the total number of frames (fraction).	
progressbar	<i>none, head, frametitle, foot</i>	none
	Optionally adds a progress bar to the top of each frame (head), the bottom of each frame (foot), or directly below each frame title (frametitle).	

3.1.4 Color theme

block	<i>transparent, fill</i>	transparent
	Optionally adds a light grey background to block environments like theorem and example .	
background	<i>dark, light</i>	light
	Provides the option to have a dark background and light foreground instead of the reverse.	

3.1.5 Font theme

titleformat title	<i>regular, smallcaps, allsmallcaps, allcaps</i>	regular
titleformat subtitle	Individually controls the format of titles, subtitles, section titles, and frame titles (see titleformat , above).	
titleformat section		
titleformat frame		

3.2 Color Customization

The included **METROPOLIS** color theme is used by default, but its colors can be easily changed to suit your tastes. All of the theme's styles are defined in terms of three beamer colors:

- **normal** text (dark fg, light bg)

- `alerted text` (colored fg, should be visible against dark or light)
- `example text` (colored fg, should be visible against dark or light)

An easy way to customize the theme is to redefine these colors using

```
\setbeamercolor{ ... }{ fg= ... , bg= ... }
```

in your preamble. For greater customization, you can redefine any of the other stock beamer colors. In addition to the stock colors the theme defines a number of **METROPOLIS** specific colors, which can also be redefined to your liking.

```
\setbeamercolor{progress bar}{ ... }
\setbeamercolor{title separator}{ ... }
\setbeamercolor{progress bar in head/foot}{ ... }
\setbeamercolor{progress bar in section page}{ ... }
```

3.3 Font Customization

The default font for **METROPOLIS** is **Fira**. This can be easily changed using the standard font selection commands of the `fontspec` package. So if you prefer, for example, the **Ubuntu** font family, just add the following two commands after loading the **METROPOLIS** theme.

```
\setsansfont{Ubuntu}
\setmonofont{Ubuntu Mono}
```

If you are expecting to present in a large room or with an underpowered projector, you may want to change the font to a heavier weight of Fira to maximize readability.

```
\setsansfont[BoldFont={Fira Sans SemiBold}]{Fira Sans Book}
```

3.3.1 Old style figures

The regular `fontspec` mechanism for changing glyph appearance applies also to this theme. If you want to have old style figures in the text but regular lined

figures for math, you could add the following to your preamble:

```
\usefonttheme{professionalfonts} % required for mathspec
\usepackage{mathspec}
\setsansfont[BoldFont={Fira Sans},
             Numbers={OldStyle}]{Fira Sans Light}
\setmathsfon(Digits)[Numbers={Lining, Proportional}]{Fira
  Sans Light}
```

3.4 Commands

3.4.1 Standout frames

The **METROPOLIS** inner theme offers a custom frame format with large, centered text and an inverted background — perfect for focusing attention on single sentence or image. To use it, add the key **standout** to the frame:

```
\begin{frame}[standout]
  Thank you!
\end{frame}
```

4 pgfplots integration

METROPOLIS comes with a set of pre-defined pgfplots styles and a color theme based on Paul Tol's color scheme.

4.1 Styles

Pass the following style keys to the axis environment to get the appropriate effect:

mlineplot Plot regular line charts with reduced axis frames, less intrusive legend and subdued grid.

mbarplot Plot vertical bar charts in a similar way as **mlineplot** but reduce grid usage.

`horizontal mbarplot` Plot horizontal bar charts.

`disable thousands separator` Helper style to remove thousands separator.

4.2 Paul Tol colors

A good presentation uses colors that are distinct from each other as much as possible as well as from black and white, can be discerned item under different lighting and display environments and by color-blind viewers, while matching well together.

In a [technical note](#) for SRON, Paul Tol proposed a palette of colors satisfying these constraints. The sub-package `pgfplots-themetol` defines palettes for `pgfplots` charts based on Tol's work.

5 Tips & Tricks

5.1 Backup Slides

Speakers will often include extra slides at the end of their presentation to refer to during audience questions. One easy way to do this is to include the `appendixnumberbeamer` package in your preamble and call `\appendix` before your backup slides.

METROPOLIS will automatically turn off slide numbering and progress bars for slides in the appendix.

6 Known Issues

6.1 Title formats

Be aware that not every font supports small caps, so the `smallcaps` or `allsmallcaps` options may not work if you use a font other than **Fira Sans**. In particular, the Computer Modern sans-serif typeface, which is used when **METROPOLIS** is compiled with pdf $\text{\LaTeX}$, does not have a small-caps variant.

The title format options `allsmallcaps` and `allcaps` are quite nice from an aesthetic point of view, but their use of `\MakeLowercase` and `\MakeUppercase` can cause unexpected problems. For example:

- Some commands, like `\`, do not work inside `\MakeLowercase` and `\MakeUppercase`. (See [#125](#))
- Only alphabetic characters are affected by `\MakeLowercase`, so numerals and punctuation remain at full height. This can spoil some of the aesthetic benefits of `allsmallcaps`. (See [#33](#))
- `\MakeLowercase` and `\MakeUppercase` apply to math mode and `\scshape` does not. This can easily introduce mathematical errors that are hard to catch.
- It is impossible to typeset symbols which are encoded as uppercase letters in a different font. In particular, `\mathbb` and `\mathcal` letters will be replaced by other math glyphs. (See [#153](#))

The `allsmallcaps` and `allcaps` options are safe to use if your titles contain only alphabetic characters and do not require the expansion of any macros.

6.2 Interactions with other color themes

METROPOLIS can be used along with any other Beamer color theme, such as `crane` or `seahorse`. If you wish to do this, it is usually best to include the **METROPOLIS** subpackages individually so the **METROPOLIS** color theme is never loaded. This will prevent conflicts between the **METROPOLIS** color theme and your preferred theme.

For example, overriding the color theme as follows may not work as expected because `\usetheme{metropolis}` loads the **METROPOLIS** color theme, which defines a relationship between the frametitle background and the primary palette of the theme. Since `seahorse` assumes a different relationship between its palettes, the result is a grey, rather than periwinkle, frametitle background.

```
\usetheme{metropolis}
\usecolortheme{seahorse}
```

The correct colors are chosen if the **METROPOLIS** outer, inner, and font themes are loaded separately:

```

\useoutertheme{metropolis}
\useinnertheme{metropolis}
\usefonttheme{metropolis}
\usecolortheme{seahorse}    % or your preferred color theme

```

Please note that **METROPOLIS** may not use all the colors defined in your favourite Beamer color theme. In particular, **METROPOLIS** does not set a background color for the title; this will cause issues when using color themes like **whale** which set a white foreground for the title.

6.3 Notes on second screen

If you use the `[show notes on second screen]` option built in to Beamer and compile with $\text{X}_{\text{E}}\text{L}_{\text{A}}\text{T}_{\text{E}}\text{X}$, text on slides following the first section slide may be rendered in white instead of the regular colour. This is due to a **bug** in Beamer or $\text{X}_{\text{E}}\text{L}_{\text{A}}\text{T}_{\text{E}}\text{X}$ itself. You can work around it either by compiling with $\text{LuaT}_{\text{E}}\text{X}$ or by adding the following code to your preamble to reset the text color on each slide.

```

\makeatletter
\def\beamer@frametnotesbegin{% at beginning of slide
    \usebeamercolor[fg]{normal text}
    \gdef\beamer@noteitems{}%
    \gdef\beamer@notes{}%
}
\makeatother

```

6.4 Standout frames with labels

Because the **standout** frame option creates a group to restrict the colour change to a single slide, labels defined after calling **standout** will stay local to the group. In other words, the following may result in a “label undefined” error.

```

\begin{frame}[standout, label=conclusion]{Conclusion}
    Awesome slide

```

```
\end{frame}
```

To fix this problem, change the order of the keys in the frame.

```
\begin{frame}[label=conclusion, standout]{Conclusion}
  Awesome slide
\end{frame}
```

This error can be unwittingly triggered if you export your slides from Emacs Org mode, which automatically adds labels after frame options. Alex Branham [offers](#) the following solution for Org mode users, using `org-set-property`.

```
* Start of a frame
:PROPERTIES:
:BEAMER_opt: label=conclusion,standout
:END:
```

6.5 Standout frames with Pandoc

With Pandoc versions prior 1.17.2 it was not possible to create standout frames because Pandoc only supported a specific list of frame attributes thus ignoring additional attributes such as `{.standout}`.

7 License

METROPOLIS is licensed under a [Creative Commons Attribution-ShareAlike 4.0 International License](#). This means that if you change the theme and re-distribute it, you must retain the copyright notice header and license it under the same CC-BY-SA license. This does not affect any presentations that you create with the theme.